

MALPRESENTATION AND MALPOSITION

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Introduction to malposition and malpresentation

➤ Important terminologies

- ❖ **Lie** - is the relationship of the long axis (spine) of the fetus to the long axis of the mother's uterus
 - The normal lie is **longitudinal**
 - Abnormal lies are; **Transverse, Oblique and variable**
- ❖ **Attitude:** is the relationship of the fetal parts to one another, (head & limb to its trunk)
 - The normal attitude is **flexion**
 - Abnormal attitude are - **extension & deflection**

Cont...

❖ **Presentation:** is the part of the fetus in the lower pole of the uterus

- The normal presentation is **vertex**
- Abnormal presentations are;
 - Breech
 - Face
 - Brow
 - Shoulder & compound

Predisposing factors

Maternal factors

- Contracted pelvis
- Pelvic tumors: uterine myomas, ovarian tumors etc
- Uterine anomalies: bicornuate uterus, uterine septum etc
- High parity

Fetal and placental factors

- Prematurity
- Fetal anomaly (e.g. hydrocephalus, anencephalus)
- Polyhydramnios / oligohydramnios
- Multiple pregnancy
- Placenta previa

Diagnostic approach

❖ Clinical assessment

- History
- obstetric palpation and
- digital vaginal examination in labor

❖ Ultrasound

- is mainly used to confirm clinical diagnosis and to investigate for predisposing factors

Malposition

❖ **Definition:** Malposition is abnormal positions of the vertex (other than occipito-anterior position) of the fetal head and mento-anterior in face presentation relative to the maternal pelvis

❖ **Classification:**

- Occiput posterior
- Persistent occiput transverse position

Cont...

❖ **Positions of vertex presentation.**

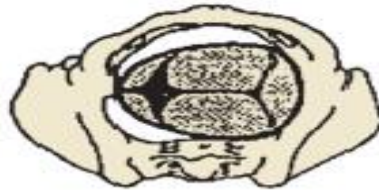
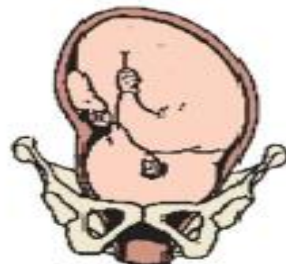
- **Left occipito anterior (LOA)** - the occiput points to the left iliopectineal eminence
- The sagittal suture is on the left oblique diameter.
- **Right occipito anterior (ROA)** - the occiput points to the right iliopectineal eminence
- The sagittal suture is in the right oblique diameter of the pelvis.

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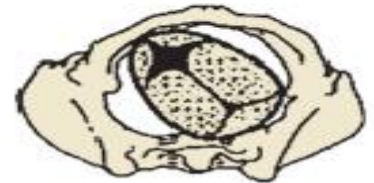
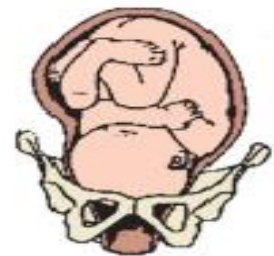
- **Left occipito lateral (LOL)** - The occiput points to the left iliopectineal line mid way between the iliopectineal eminence & the sacroiliac joint
- The sagittal suture is in the transverse diameter
- **Right occipito lateral (ROL)** - The occiput point to the right iliopectineral line midway between iliopectineal eminence & sacro iliac joint
- The sagittal suture is in the transverse diameter of the pelvis.
- **Left occiputo posterior (LOP)** - The occiput points to the left sacroiliac joint. The sagittal suture is in the left oblique diameter of the pelvis
- **Right occiputo posterior(ROP)**- The occiput points to the right right sacroiliac joint
- The sagittal suture is in right oblique diameter of the pelvis.



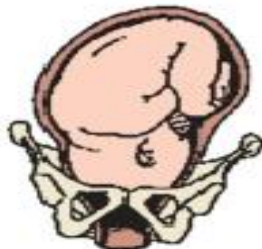
LOA



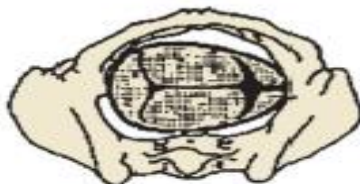
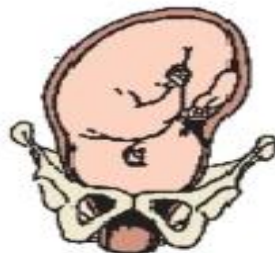
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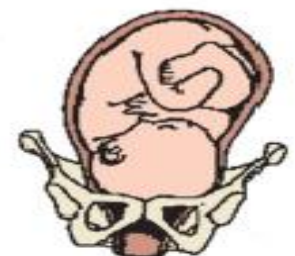
LOP



ROA



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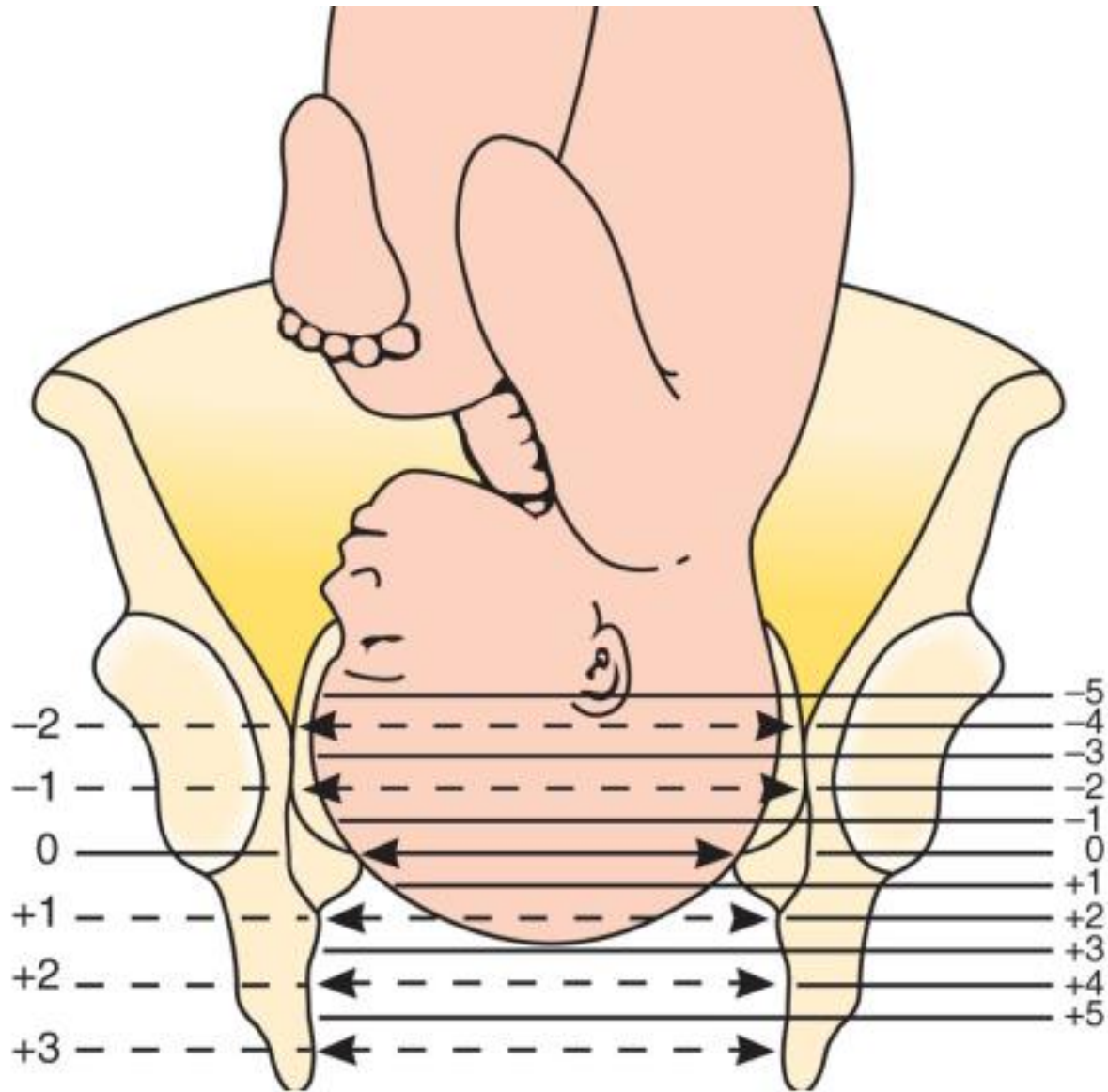


ROP

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- **Denominator:** The part of the fetus which determines the position of the presenting part which is used when referring the fetal position
 - Vertex - Occiput
 - Breech - Sacrum
 - Face –mentum
 - Brow- sinciput
- **Engagement** - When the biparetal diameters of the fetal head passes through the pelvis brim
- **Station**-distances of the fetal presenting part from the maternal ischial spine

Cont....



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➤ **Occiput posterior position:**

- ✓ Occipito-posterior position is when the fetal occiput is posterior in relation to the maternal pelvis

❖ **Diagnosis**

❑ **Suggestive abdominal findings:-**

- Flattened lower part of the abdomen
- Anteriorly palpable fetal limbs. o Fetal heart heard in the flank

❑ **On vaginal examination:-**

- Posterior fontanelle towards the sacrum
- Anterior fontanelle felt anteriorly if neck is flexed

Cont...

➤ **Management**

- If there is rotation to occiput anterior, expect vaginal delivery as occiput anterior
- If there is arrest of fetal descent at low station (station at or below +2), try **Vacuum delivery**
- Deliver by **C/S** if there is Arrest of fetal descent(station above +2) in the presence of adequate uterine contraction

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❖ **If no rotation:**

- Expect vaginal delivery as long as fetal condition is stable, maternal pelvis is adequate and there is progressive descent of the fetus
- Augment with oxytocin if there is no contraction
- Try vacuum delivery in case of 2nd stage abnormality
- Deliver by C/S if there is arrest of fetal descent and the prerequisites for vacuum delivery are not met

Cont...

➤ **Persistent occiput transverse position**

❖ **Definition:** Persistent occiput transverse position is an occiput transverse position that is maintained for **1 hour** or more in the second stage of labour

❖ **Classification**

- High transverse arrest (arrest above station +2)
- Deep transverse arrest (arrest below station +2)

Cont...

❖ **Diagnosis**

- Protracted or arrested fetal descent

□ **On vaginal examination**

- the fetal sagittal suture and fontanelles are palpable in the transverse diameter of the pelvis
- the fetal ears can be palpated superiorly under the symphysis and inferiorly above the sacrum/coccyx
- There may be anterior or posterior asynclitism

Cont...

➤ Management

- If there is any progress in descent and the fetal heart rate is reassuring, expectant management is the preferred option
- Augmentation
- Cesarean delivery if there is high transverse arrest despite adequate uterine contraction and maternal expulsive effort

Malpresentation

- ✓ Malpresentations are all presentations other than vertex
- ✓ Types of malpresentation;
 - Breech
 - Face
 - Brow
 - Shoulder and
 - Compound

BREECH PRESENTATION

- ✓ Breech presentation is when the fetal **buttock and/ or feet** occupies the lower part of the uterus
- ✓ The fetal head occupies the fundus
- The lie is longitudinal
- The denominator is the sacrum for frank, complete & footling breeches
- The presenting diameter is bitrochantric which is 10 cm

➤ **Fetal position in breech presentation**

- Determined by using the fetal **sacrum** as the point of reference to the maternal pelvis
- Eight possible positions are recognized:
 - sacro-anterior (SA)
 - sacro-posterior (SP)
 - left sacro-transverse (LST)
 - right sacro-transverse (RST)
 - left sacro-anterior (LSA)
 - left sacro-posterior (LSP)
 - right sacro-anterior (RSA), and right sacro-posterior (RSP)

Cont...

❖ Types of Breech presentation

1. Frank breech

- Both hips are flexed
- Both knees are extended
- Neither of the feet is felt
- Both ischial tuberosities sacrum and anus are palpable
- 60-65%



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2. Complete or flexed breech

- Both hips and knees are flexed
- feet are felt above the buttock or alongside the buttock
- Anal orifice and ischial tuberosities felt above the buttock
- About 5%



Cont..

3. Footling breech /incomplete breech

- one or both hips are extended with one or both knee extended
- One or both feet are felt inferior to the buttock
- 10-40%

Cont...

Single footling breech



Frank breech



Complete breech

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Cont...



**Complete
breech**



**Incomplete
breech**



**Frank
breech**

Cont..

4. Kneeling breech;

- one or both hip(s) are extended with flexed knee(s)
- The knee(s) felt first bellow the breech

➤ **Diagnosis of breech presentation**

❖ **History**

- Discomfort under the rib
- Feeling of more fetal movement in the lower segment

❖ **Abdominal examination**

- Head is felt as hard, globular and ballotable mass at the fundus
- Breech is felt soft and bulky at the lower pole of the uterus
- FHB auscultate above umbilicus

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❖ **Vaginal examination**

❖ **Ultrasound:** To confirm clinical diagnosis

- To assess fetal attitude
- To assess amniotic fluid volume to consider ECV
- To estimate fetal weight, and to investigate for fetal anomalies

Cont...

➤ **Antepartum management**

- For less than 36weeks;
 - Follow closely to evaluate for spontaneous version to cephalic presentation
 - ultrasound for possible anomalies
- For more than 36weeks; consider **ECV**

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➤ **ECV (External Cephalic Version)**

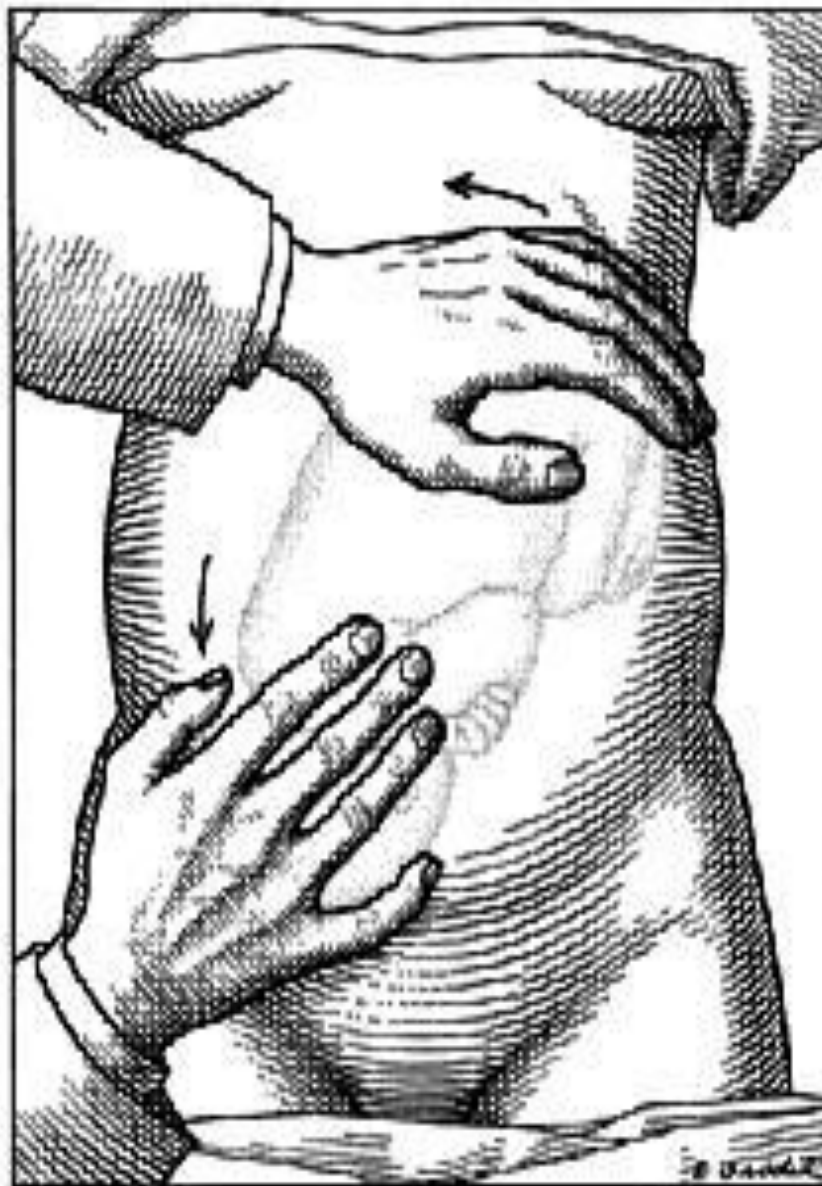
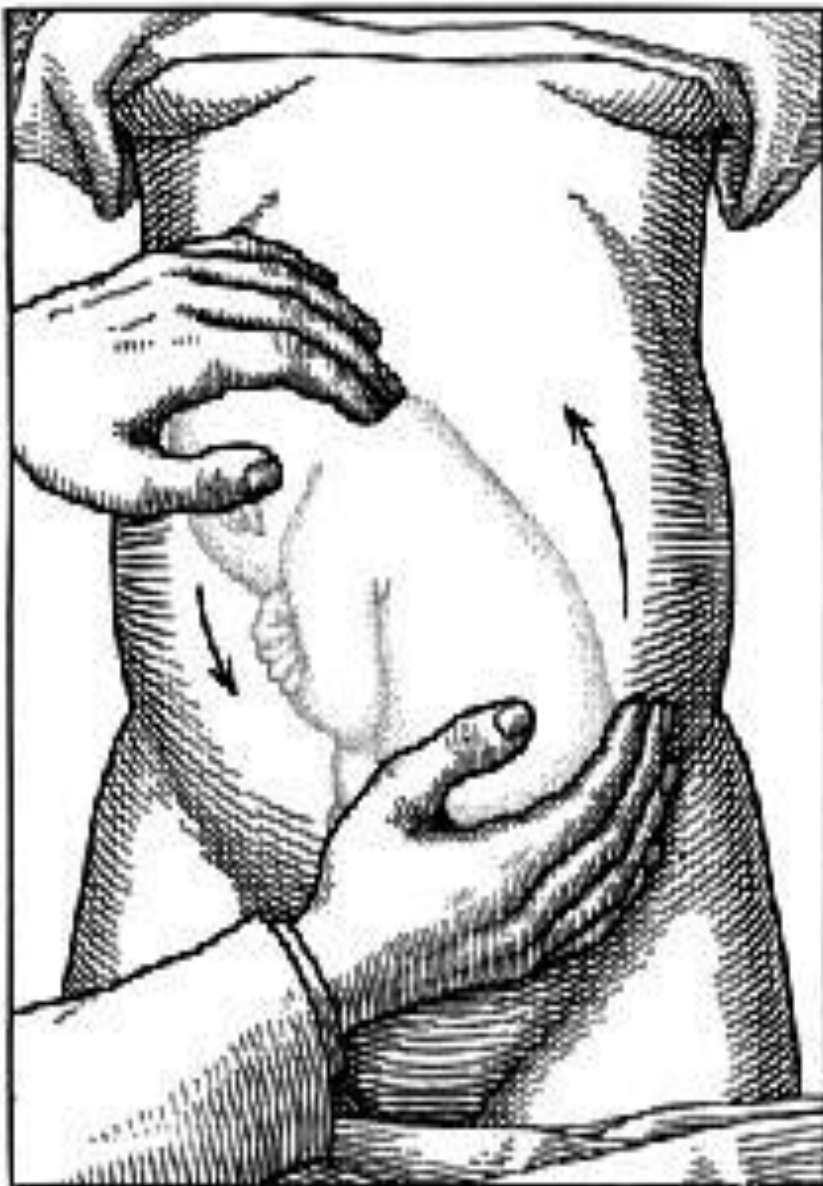
- ECV is a procedure in which the baby is manipulated by pressure through the mother's abdominal wall into a cephalic presentation
- It is a third alternative to vaginal delivery or cesarean delivery for the breech fetus
- Performed during antenatal near to term so as to increase chance of vertex delivery
- Success with ECV varies from 60 to 75 percent

❖ **Indications of ECV**

- Breech presentation
- Transverse lie and Oblique lie

Contraindications to ECV

- placenta previa
- Uterine and membrane rupture
- fetal distress
- Hyperextended fetal head
- Abruptio placentae
- Footling breech
- Previous CS
- Multiple pregnancy
- Polyhydramnios
- Oligohydramnios
- PROM
- IUGR
- Congenital fetal abnormality
- Rh-immunization
- Maternal cardiac disease, Hypertension
- Cord completely encircling the fetal neck
- Preterm labor



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➤ **Prerequisites of ECV**

- Gestational age > 36 weeks including early labor
- Intact fetal membranes and adequate amniotic fluid
- Reassuring fetal condition
- No contraindication for vaginal delivery

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➤ **Failed ECV:**

- Version attempts should be discontinued, if there is
 - Excessive maternal discomfort
 - Persistent abnormal FHR
 - After a maximum of three attempts

Cont...

- **Complications of ECV**
 - Abnormal FHR usually transient
 - Feto-maternal hemorrhage
 - Vaginal bleeding and placental abruption
 - Fracture of the baby's femur

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➤ **Mechanism of labour in breech delivery**

- Engagement occur when the bitrochanteric diameter of the fetus has passed the plane of the maternal pelvic inlet
- At the levatorani muscular sling, internal rotation brings the bitrochanteric diameter into the anteroposterior axis of the pelvis
- The breech at the outlet emerge, first as a sacrum transverse, then rotating to sacrum anterior position

➤ **Management of vaginal breech delivery**

❖ **First stage of labour**

- Monitor using partograph and resuscitate mother
- Consider analgesics as labor pain management
- Immediate vaginal examination at rupture of membranes to rule out cord prolapse
- Instruct mother not to push until the cervix is fully dilated
- Augmentation of labor is contraindicated in breech presentation
- Cesarean section if cervicogram crossed the action line
- For second stage of labour use the following techniques of vaginal breech extraction

➤ **Techniques of vaginal breech delivery**

❖ **Spontaneous breech delivery**

- the entire fetus is delivered spontaneously without manipulation than supporting the fetus

❖ **Assisted breech delivery (partial breech extraction)**

- the fetus is delivered spontaneously as far as the umbilicus
- After this, delivery of the legs, trunk, and arms are assisted manually with operator traction and assisted maneuver
- the head may be delivered manually or with forceps

❖ **Total breech extraction**

- The entire fetus is manually delivered
- Nowadays ,It is only done for second twin in breech presentation





➤ **Criteria for vaginal breech delivery**

- No contraindication to vaginal delivery
- Gestational age 36 weeks or more
- Absence of fetal anomaly
- Frank or complete breech
- EFW < 3500 gm
- Normal progress of labor
- Flexed fetal head(Confirmed by Ultrasound)
- Continuous Fetal heart monitoring available
- Staff skilled in breech delivery
- Available facility for emergency C/S

➤ **Pinnard's maneuver**

- Used for delivery of leg

❖ **procedure**

- Wait until body is born to the level of the umbilicus
- Use a dry towel to wrap around the hips to help with gentle traction of the infant)
- Put Two fingers along one extremity to the knee,
 - then push away from midline which causes flexion and subsequent delivery of the lower leg
- Do not hold the baby by the flanks or abdomen as this may cause kidney or liver damage

➤ **Lovset's maneuver**

- Used for delivery of arm

❖ **procedure**

- When extended arms fail to deliver on down traction ,
- Hold the fetus around the bony pelvis with thumbs across the sacrum
- The fetus is turned through half a circle (180 degree) while downward traction is applied at the same time
- then the posterior arm emerges under pubic arch
- The position is restored and anterior arm is delivered in the same manner

➤ **Mauriceau Smellie Veit Maneuver (MSV)**

- Used for delivery the after coming head

❖ **procedure**

- Index and middle finger of one hand are applied over the maxilla, to flex the head, while the fetal body rests on the palm of the same hand and forearm
- Fetal legs straddle the forearm
- Two fingers of the other hand are hooked over the fetal neck and grasp the fetal shoulders
- Apply gentle downward traction to deliver the head
- Piper forceps) can be used if the MSV cannot be accomplished easily

➤ **Cesarean delivery of breech**

- ✓ Cesarean delivery of breech is indicated when;
 - EFW <1,500 or >4,000 g
 - Footling presentation
 - Small pelvis
 - Hyperextended fetal head
 - placenta previa
 - Absence of expertise for vaginal breech delivery
 - Non-reassuring fetal heart rate pattern
 - Arrest of progress

FACE PRESENTATION

➤ Definition

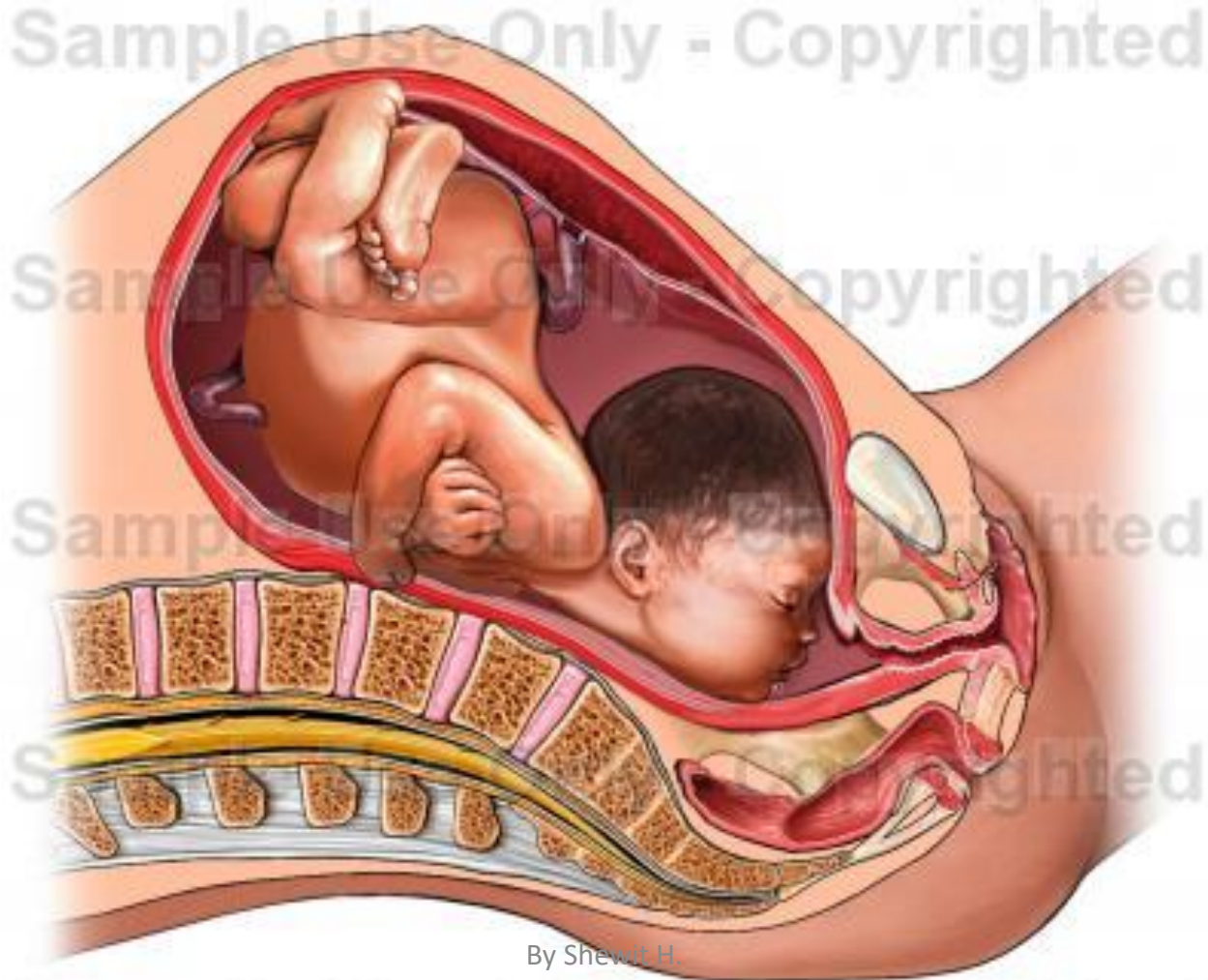
- Face presentation is hyperextension of the fetal head with the face being the leading part
 - Fetal chin (mentum) is used as a reference point

➤ Diagnosis

❖ Suggestive abdominal finding:

- Groove may be felt between the occiput and the back
- The presenting diameter is submentobregmatic which is 9.5cm
- Attitude is hyperextension
- Incidence:- 1 in 500 live births

Cont...



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Cont...

❖ vaginal examination:

- Fetal chin, mouth and nose are palpated
- The mouth with the two malar bone prominences make a triangle

Cont...

➤ Management

❖ Mento-anterior:

- follow the progress of labor in grossly adequate pelvis
- Augmentation is not recommended
- Forceps delivery can also be used
- Vacuum delivery is contraindicated

Cont...

❖ Mento-posterior:

- Follow labor progress with anticipation of vaginal delivery in early admissions
- In persistent mento-posterior presentation;
 - If fetus is alive → Cesarean delivery
 - If the fetus is dead → Craniotomy if all the prerequisites are met

BROW PRESENTATION

➤ Definition

- Partial extension of the fetal head making the occiput higher than the sinciput
- Brow (Frontal bone) is the presenting part (the part of the head between orbital ridges and anterior fontanel)
- Engagement is usually impossible and arrested labor is common
- The presenting diameter is mentovertical diameter- 13.5 cms

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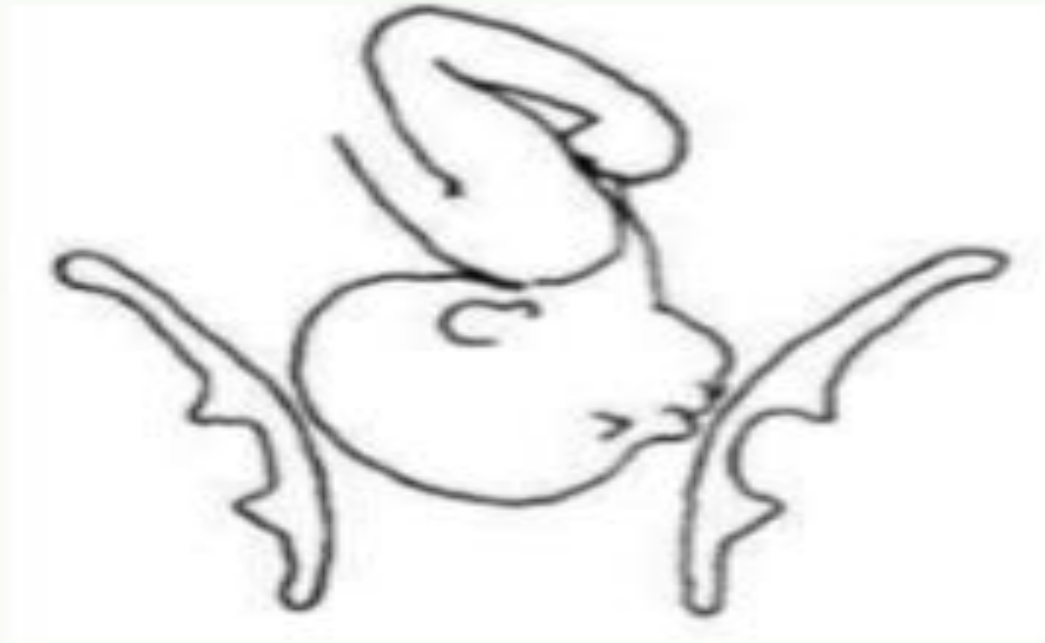


Figure 9. Brow presentation

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➤ **Diagnosis**

- Occiput felt above sinciput during abdominal examination
- Anterior fontanelle, coronal suture and orbits are felt on vaginal examination

Cont...

➤ Management

- In case of persistent brow presentation;
 - deliver by cesarean section if the fetus is alive
 - Perform craniotomy if the fetus is dead
- Deliver by cesarean section if:-
 - the cervix is not fully dilated or station is high
 - Do not use an obstetric vacuum or forceps
 - Augmentation of labor is also not generally recommended
 - vaginal delivery is impossible

SHOULDER PRESENTATION

➤ Definition

- Shoulder presentation is when the long axis of the fetus is perpendicular to the longitudinal axis of the mother/uterus
- The shoulder is the presenting part in a transverse lie

➤ Diagnosis

➤ On abdominal examination;

- Neither the fetal head nor the breech is felt in the upper and lower part of the uterus
- The abdomen is transversely elongated than longitudinally
- Fundal height is less than gestation age
- The shoulder or the prolapsed arm is felt on vaginal examination

Cont...



Cont...

➤ Management

- CS delivery if a transverse lie presents in labor in a term pregnancy
- Consider ECV during pregnancy after 36 weeks of gestation until early in labor with intact fetal membranes,
- If ECV fails deliver by CS

COMPOUND PRESENTATION

➤ Definition

- Compound presentation is when a fetal extremity prolapses alongside the main presenting part
- It usually is the hand alongside the fetal head

➤ Diagnosis

- Irregular mobile fetal part adjacent to the larger presenting part on vaginal examination

Cont...

➤ Management

- Closely monitor labor
- The prolapsed extremity should not be manipulated
- Spontaneous vaginal birth can occur only when the fetus is very small or dead and macerated
- Cesarean delivery is indicated if there is protraction or arrest of labor
- Augmentation of labor is not recommended.

THANK YOU